“What You Optimize Matters: The Specification Problem in Operational Optimization”

You cannot optimize your way out of a bad problem specification

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THE SPECIFICATION PROBLEM

Research Question and the setting- How do pre-optimization specification choices shape outcomes?

- Every optimization project makes two decisions: what problem to solve, and how to solve it. We have rigorous methods for choosing algorithms, but the problem specification often gets much less scrutiny
- Objectives, thresholds, and operating rules are commonly set before the solver runs, even though they can strongly shape the results
- This study asks a simple question: **What if we treated those specification choices as decisions to test, not inputs to accept?** Using real operational data, we compare the impact of specification choices with algorithm choice across 55,080 runs
- The results put the focus upstream: **Before asking how to optimize, check whether the problem has been specified well.**

North American timber and building products with dedicated trucking program

Real weekly shipment orders and driver records, no synthetic instances added

Clarke Wright savings heuristic with home anchor, it is held constant across all runs

Seven pre-solver specification parameters, fully crossed across 55080 runs

Identification Strategy:
Keeping the algorithm fixed means every difference in outcome traces to specification choices alone, with no algorithmic confound

EXPERIMENTAL DESIGN

Separating specification effects from algorithm effects.

$$\text{Maximize } \sum S_w(i,j)\cdot x_{ij} \quad \text{where} \quad S_w(i,j) = w(i,j)\cdot[c(d,i)+c(d,j)-c(i,j)], \ \text{summed over eligible pairs in the candidate pool}$$

Notation	Parameter	Description	No. of Levels	Values / Levels Used
θ_F	Floor Threshold	Minimum guaranteed earnings floor per week ($). Route cost = max(RATE × total_miles, floor), where RATE = $1.50/mile.	17	$1,800 – $5,000 in $200 steps (reference: $4,000)
θ_P	Proximity Radius	Maximum road distance (miles) from the carrier's home terminal to an order's destination. Any order whose delivery destination falls within that radius, gets force-included in the carrier’s candidate pool regardless of how it scores on cost or deliverability, and it becomes eligible to be the anchor (the mandatory final stop on the route)	9	50, 75, 100, 125, 150, 175, 200, 225, 250 miles (reference: 100)
θ_N	Pool Size	Number of top-ranked standard orders included in each carrier's candidate pool. Orders are scored on a weighted composite of open-market cost, origin proximity to carrier home, and deliverability. Difficult-lane and anchor-eligible orders are force-included regardless of this cap.	5	150, 200, 250, 300, 350 orders (reference: 200)
θ_W	Weight Scheme	Weights (w_cost, w_proximity, w_deliverability) applied to the composite scoring function that ranks standard orders for pool inclusion. These weights govern WHICH orders enter the candidate pool and are not applied to the Clarke-Wright savings formula.	4	equal (0.33, 0.33, 0.34) cost_heavy (0.60, 0.20, 0.20) proximity_heavy (0.20, 0.60, 0.20) delivery_heavy (0.20, 0.20, 0.60) (reference: equal)
θ_A	Anchor Criterion	Rule for selecting the mandatory final stop (anchor) of each route from orders whose destination falls within the proximity radius of the carrier home. The anchor is a forced last leg ensuring the driver returns near home (C5).	2	highest_omv — highest open-market value order longest_leg — longest loaded-mile order (reference: highest_omv)
θ_O	Objective Function	Scoring criterion used to rank and select among all feasible routes generated for a carrier. Only the top-ranked route per carrier is retained. cost_min maximises total savings; coverage_max maximises orders served; efficiency_max maximises the loaded-mile ratio.	3	cost_min: maximise total savings ($) coverage_max: maximise orders served efficiency_max: maximise loaded-mile ratio (reference: cost_min)
θ_R	Retention Rule	Filter applied after full feasibility check to decide which routes are retained. positive_savings keeps only routes that beat open-market cost; all_feasible keeps every constraint-satisfying route; efficiency_threshold keeps routes with loaded efficiency ≥ 60%.	3	positive_savings — savings > $0 all_feasible — any feasible route efficiency_threshold — loaded eff. ≥ 60% (reference: positive_savings)

17 * 9 * 5 * 4 * 2 * 3 * 3 = 55080 runs

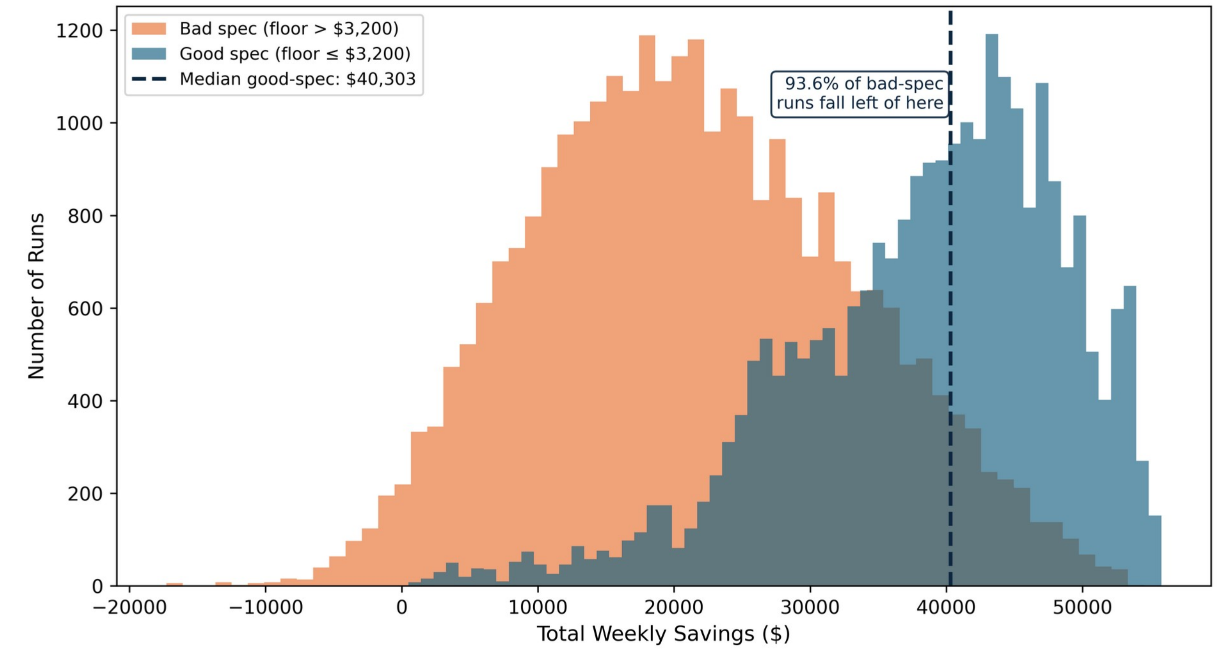
KEY FINDINGS (MAIN EFFECTS)

“Renegotiating one contract clause can move an operation into an entirely different regime of savings”

~ 94%
of bad-spec runs fall below the median of good-spec runs

1.84x
More savings per driver under good-spec vs bad-spec

32%
Of the max possible weekly saving is unlocked by moving across regimes



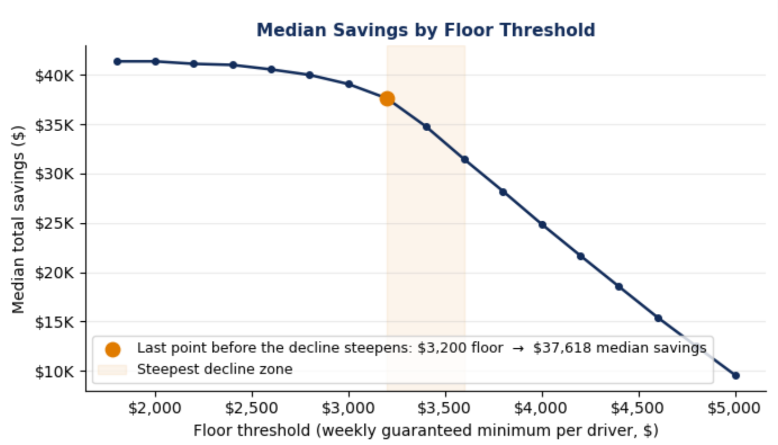
“ Weekly savings is the difference between what a set of orders would have cost on the open market and what it costs to run them as one dedicated weekly route, where route cost is the greater of standard mileage pay or the guaranteed floor.
Total weekly savings sums this difference across every driver assigned a route that week.”

$$\text{Savings}_d = \sum_{o \in R_d} \text{OMV}(o) - \max(\text{RATE}\cdot m_d,\ \theta_F)$$

$$\text{Total Weekly Savings} = \sum_{d \in D} \text{Savings}_d$$

where R_d = orders on driver d’s route, m_d = total route miles, θ_F = floor threshold, D = drivers assigned a route

Why $3200: Where the Decline Steepens



The Practitioner’s Implication

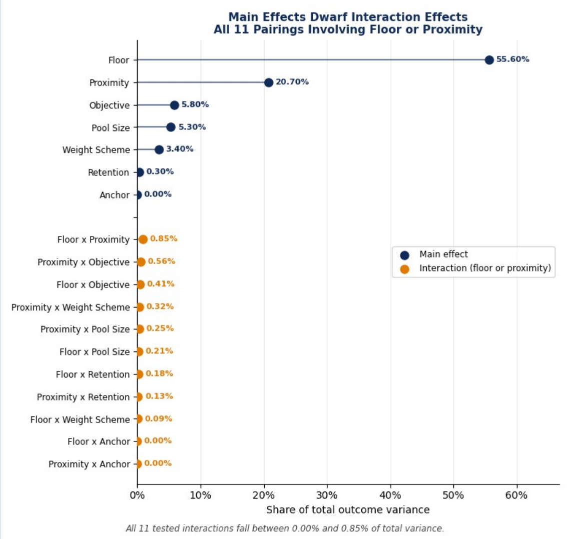
The 3200 floor is not an arbitrary cut-off is the point where marginal savings recovery is highest per dollar of floor reduction.

Negotiating below 3200 still gains savings, but at a diminishing rate.

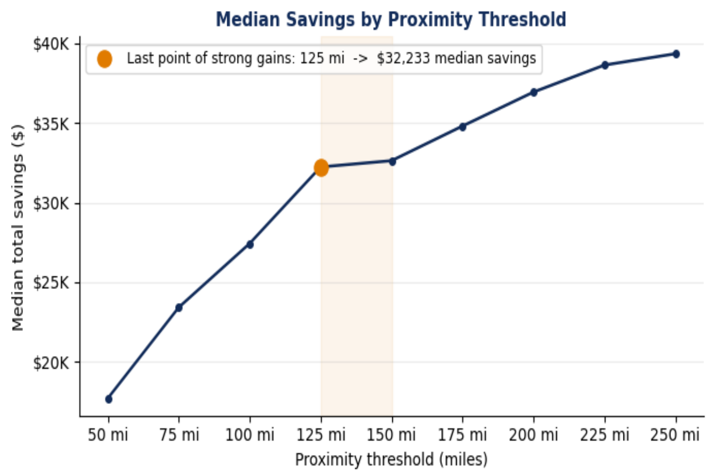
Negotiating above it, especially in the 3200–3600 high-damage zone costs disproportionately.

KEY FINDINGS

“Not all parameters matter equally, and because interaction effects are negligible, each one delivers its impact independently”



All 11 tested interactions fall between 0.00% and 0.85% of total variance.

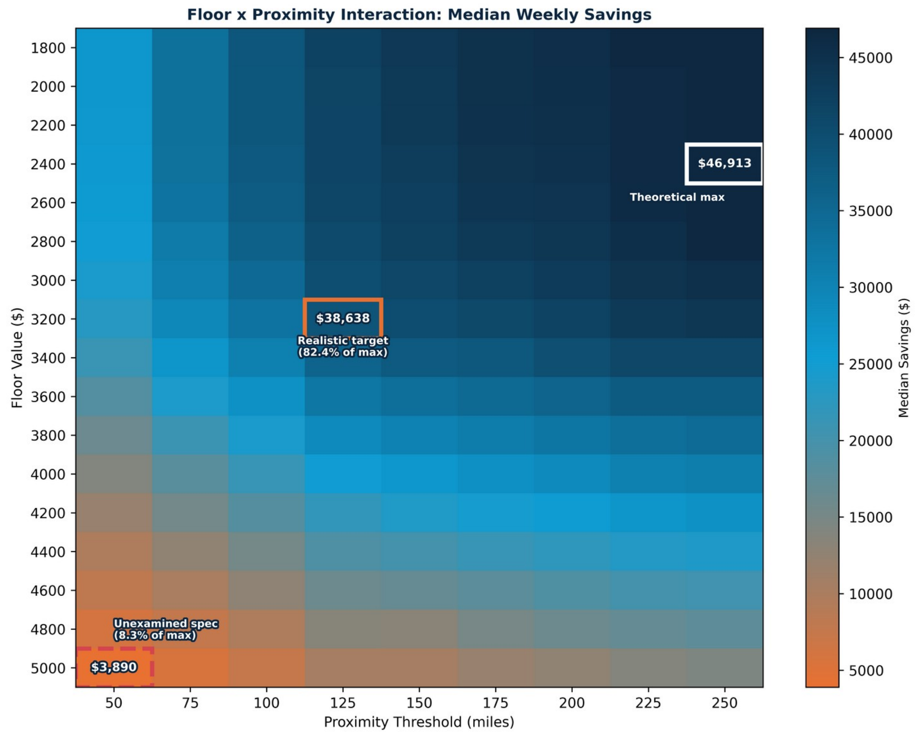


The Practitioner’s Implication

The floor and proximity thresholds are separate negotiation levers. Without significant interaction effects, savings from one doesn’t depend much on the other parameter. We can hence pursue each negotiation on its own terms and expect the gains to add up.

Gains by setting a proximity threshold is strong and consistent up to 125 miles, capturing most of the lever’s value, before the curve flattens.

“The Realistic Target: 82.4% of the Maximum”



$3200
Floor threshold at realistic negotiation target

125 mi
Proximity radius at realistic negotiation target

82.4%
Maximum Savings that can be unlocked

“Negotiating floor from $5,000 to $3,200 and proximity from 50 to 125 miles increases median weekly savings by roughly 10x, a gain achieved entirely through specification choices, before any algorithmic optimization is applied.”

THE DECISION BEFORE THE SOLVER

A bi-level formulation with a pre-optimization specification decision that the solver inherits

Upper level – Specification selection

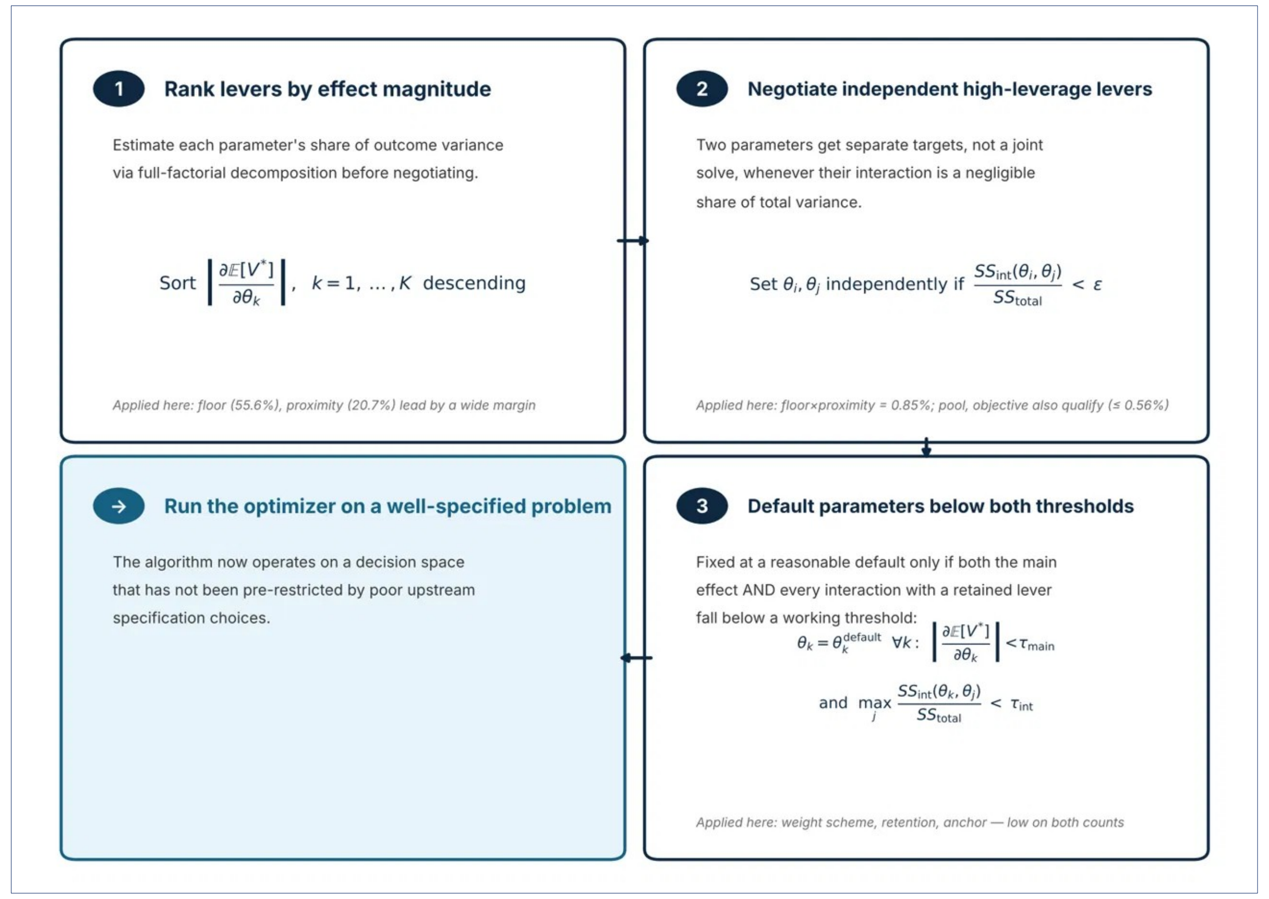
$$\theta^* = \arg\max_{\theta \in \Theta} \ \mathbb{E}[\,V^*(\theta)\,]$$

$$\theta = (\theta_F,\ \theta_P,\ \theta_N,\ \theta_W,\ \theta_A,\ \theta_O,\ \theta_R)$$

Lower level – Routing given θ

$$V^*(\theta) = \max_x \sum S_w(i,j)\cdot x_{ij}$$

$$\text{s.t.}\quad i \in O(\theta_F,\theta_P),\quad |R_k| \le \theta_N \ \ \forall k$$



① **Rank levers by effect magnitude**

Estimate each parameter's share of outcome variance via full-factorial decomposition before negotiating.

$$\text{Sort } \left|\frac{\partial \mathbb{E}[V^*]}{\partial \theta_k}\right|, \quad k = 1, \ldots, K \ \text{ descending}$$

Applied here: floor (55.6%), proximity (20.7%) lead by a wide margin

② **Negotiate independent high-leverage levers**

Two parameters get separate targets, not a joint solve, whenever their interaction is a negligible share of total variance.

$$\text{Set } \theta_i, \theta_j \text{ independently if } \frac{SS_{\text{int}}(\theta_i,\theta_j)}{SS_{\text{total}}} < \varepsilon$$

Applied here: floor×proximity = 0.85%; pool, objective also qualify (≤ 0.56%)

③ **Default parameters below both thresholds**

Fixed at a reasonable default only if both the main effect AND every interaction with a retained lever fall below a working threshold:

$$\theta_k = \theta_k^{\text{default}} \quad \forall k : \left|\frac{\partial \mathbb{E}[V^*]}{\partial \theta_k}\right| < \tau_{\text{main}}$$

$$\text{and} \quad \max_j \frac{SS_{\text{int}}(\theta_k,\theta_j)}{SS_{\text{total}}} < \tau_{\text{int}}$$

Applied here: weight scheme, retention, anchor — low on both counts

→ **Run the optimizer on a well-specified problem**

The algorithm now operates on a decision space that has not been pre-restricted by poor upstream specification choices.

FUTURE WORK: SPECIFICATION UNDER UNCERTAINTY

Specifications are generally policy decisions which are set before a full picture of demand is known

The $3,200 / 125mi target above is a best response to realized demand trace where every one of the 55,080 runs shares the same fixed dataset.
In practice, floor and proximity are policy parameters locked in before the complete demand they will face is known and will not have the luxury of knowing the demand before-hand

$$\theta^* = \arg\max_{\theta \in \Theta} \ \mathbb{E}\big[V^*(\theta) \mid \text{demand} \sim F(\cdot)\big]$$

$$\text{s.t.}\quad \mathbb{P}\big(V^*(\theta) \ge V_{\min}\big) \ge \alpha$$

This replaces a single realized outcome with its expectation over a demand distribution and requires the chosen specification to clear a savings bar with probability α, correcting for the overfitting risk in the deterministic result above.

The specifications are decided before the algorithm runs and, in most operations, before demand is even known. If we get that decision right under uncertainty, the solver does the rest.

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